Node class	num of nodes	cap. per node	activity factor
local logic outputs	20000	3	0.12
medium wires	4000	18	0.08
long global wires	600	95	0.03

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Supply V	0.9	
Clock Freq	1.2	Ghz

1A

Pdyn = sum(Ci*V^2*f*activity factor)

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs	20000	3	0.9	1.2	0.12	**6998.4**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.03	**1662.12**

The local logic output class dominates.

The answer is not obvious from capacitance alone; I would have guessed that the long global wires would have had the highest Dynamic Power, because they had the highest capacitance per node and Pdyn is proportional to the capacitance per node, but the other factors were large enough that the answer was not obvious simply from the capacitance values.

1B

Total dynamic switching power: sum of all dynamic power in circuit.

local logic outputs	6998.4
medium wires	5598.72
long global wires	1662.12

sum = 14259.24

Total dynamic switching power: 14259.24 equals 14.3mW

total short circuit power: **8% of total dynamic switching power.**

total dynamic switching power= 14259.24

8% total dynamic switching power= 1140.7392

total short circuit power: **1140.7392** 11.4mW

1C

leakage current per minimum-equivalent transistor 2.5 nA

num of transistors 1800000

Total leakage current: leakage per transistor * num of transistors 4500000

Power = Current * Voltage, P=IV V=0.9

Total leakage power = **4050000** **equals** **4.05 mW**

Dynamic switching power: 14259.24 equals 14.3mW

(from previous question)

Total leakage power is far lower than the dynamic switching power.

1D

Old design:

Node class	num of nodes	cap. per node	activity factor
local logic outputs	20000	3	0.12
medium wires	4000	18	0.08
long global wires	600	95	0.03
Supply V	0.9		
Clock Freq	1.2	Ghz	

Pdyn = sum(Ci*V^2*f*activity factor)

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs	20000	3	0.9	1.2	0.12	**6998.4**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.03	**1662.12**
					sum Pdyn:	14259.24

New design:

Node class	num of nodes	cap. per node	activity factor
local logic outputs 1	20000	3	0.12
local logic outputs 2	5000	3	0.05
medium wires	4000	18	0.08
long global wires	600	95	0.03

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs 1	20000	3	0.9	1.2	0.12	**6998.4**
local logic outputs 2	5000	3	0.9	1.2	0.05	**729**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.015	**831.06**
					sum Pdyn:	14157.18

Yes, this change would reduce total dynamic power as shown above. The old Pdyn total was 14259, and the new value is 14157.

1E

The student is wrong, but I cannot prove them wrong with my numerical results. At all points, the local logic output node class use the most dynamic power in total, but if the other factors (capacitance, activity factor, etc. were different, then there is no reason why another node class couldn't dominate.

	Supply V (V)	1					
	activity factor	0.15					
	Frequency	800MHz					
	Case	Input trans. time	Load Cap.	Peak short-circ. current	short-circ pulse width		
	A	20	20	90	18		
	B	80	20	210	70		
	C	80	200	95	55		
		(ps)	(fF)	(microA)	(ps)		
2A	short circuit energy per transition:						
	P = IV *t						
	however, since the pulses are triangular, area of triangle = 0.5*h*w						
	therefore:						
	P = 0.5* IV*T						
	Case	Peak short-circ. current	short-circ pulse width	**Short Circuit Energy per transition**			
	A	90	18	**810**	**0.81fJ**		
	B	210	70	**7350**	**7.35 fJ**		
	C	95	55	**2612.5**	**2.6125fJ**		
		(microA)	(ps)				
	Average short circuit power:			Note: shape triangle.			
	1/2 * i*v *t*f						
2B	Case	Peak short-circ. current	Voltage V	T time ps	F	Avg. Short Circuit Power	**Avg. Short Circuit Power**
	A	90	1	18	800	648000	**648nW**
	B	210	1	70	800	5880000	**5880nW**
	C	95	1	55	800	2090000	**2090nW**
		(microA)		(ps)	MHz		
	Case	Input trans. time	Load Cap.	short-circ pulse width			
	B	80	20	70			
	C	80	200	55			
		(ps)	(fF)	(ps)			
2C	B and C have the same input transition time but different load capacitance. A larger load can reduce the Short Circuit Current Pulse time, as it requires more to actually change value and output the stored charge and so the output value is closer to Vdd or 0. I tend to imagine it like water from a fire hose? It requires a lot of energy to turn on, so the circuit spends less time with both NMOS and PMOS passing energy and a lot more time closer to the 'ideal' values.						
	This happens even if capacitive switching energy increases because that's a whole different mechanism. Switching energy is related to the amount of energy needed to switch the state and is only related to Cl and Vdd. it isn't related to the current pulse time at all.						

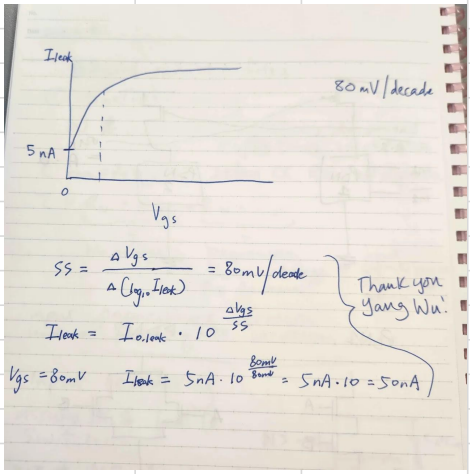
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	Case	Input trans. time	Load Cap.	Peak short-circ. current	short-circ pulse width		
	A	20	20	90	18		
	C	80	200	95	55		
		(ps)	(fF)	(microA)	(ps)		
	Supply V (V)	1					
	activity factor	0.15					
	Frequency	800MHz					
	E (low to high)	equals	Cl Vdd^2				
2D	Case	Load Cap.	Supply Voltage	**E Low to High**			
	A	20	1	**20**			
	C	200	1	**200**			
		(fF)		**(fJ)**			
	Case	Short Circuit Energy per transition					
	A	0.81fJ					
	C	2.6125fJ					
	In both cases A and C, the short circuit energy per transition is far less than the capacitive switching energy. Case C's short circuit energy is comparatively much smaller, proportional to the ordinary capacitive switching energy, than Case A.						
	Option 1:						
	Reduce the current flowing when the circuit is switching values and short circuits. This would cause lower current values in the circuit overall, though, which would not be good because the circuit's speed would dramatically decrease. This could also be done by lowering Vdd to double Vth, to lower the voltage that surpasses the threshold voltage.						
2E							
	Option 2:						
	Change the gates! Use transistors with a high Vth value to reduce the time that short circuiting could occur. A drawback for this is that values take longer to propagate through the circuit which also is not good.						

3A

Subthreshold slope		80mv/decade
Vgs	Leakage current	
0	5nA	



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SS = Subthreshold Slope

Voltage Vgs	Original I Leak	SS	**Leakage current**	Nominal Case
0	5	80	**5**	
80	5	80	**50**	
160	5	80	**500**	
240	5	80	**5000**	
mV	nA	mV/decade	nA	

3B

Vds = Vgs-Vth.

If Vth is lowered by 80mV, and the effective value of Vds remains the same, then Vgs is effectively increased by 80mV. This is because the leakage current formula depends on the difference between Vth and Vgs, not strictly Vgs (?).

Voltage Vgs	Added 80mV	Original I Leak	SS	**Leakage current**	Threshold Shift
0	80	5	80	**50**	
80	80	5	80	**500**	
160	80	5	80	**5000**	
240	80	5	80	**50000**	
mV	mV	nA	mV/decade	nA	

3C

Voltage Vgs	Original I Leak	SS	Leakage current	Nominal Case
0	5	80	5	
mV	nA	mV/decade	nA	

50000	devices
250000	total block current
0.9	supply voltage
225000	Power
2.25mW	

Voltage Vgs	Added 80mV	Original I Leak	SS	Leakage current	Threshold Shift
0	80	5	80	50	
mV	mV	nA	mV/decade	nA	

	50000	devices
	2500000	total block current
	0.9	supply voltage
	2250000	Power
	22.5mW	
3D	A designer might intentionally use lower threshold devices on a critical path despite the leakage penalty to reduce the propagation delay from changing inputs. This is important because a critical path, the longest path possible, is necessarily the bottleneck for maximum speed of the circuit, and so reducing this path by any means necessary benefits the block as a whole.	
3E	This is absolutely not correct - the threshold shift changes the leakage by a power of 10 with each shift of 80mV! at tiny voltages, that might be small, but the change in leakage grows so quickly.	

4A

L	R/unit length	C/unit length	Intrinsic date delay	
500	0.08	0.2	35	
2000	0.08	0.2	35	
5000	0.08	0.2	35	
micrometer	ohms/micrometer	fF/micrometer	ps	

Wire delay = 1/2*RLCL

L	RL	CL	Wire delay	
500	40	100	2000	2ps
2000	160	400	32000	32ps
5000	400	1000	200000	200ps
			fs	

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4B

gate delay = 35ps

35ps =	0.5*RCL^2	
35ps*2/RC=	L^2	
4375000	equals L^2	(calculating with femtoseconds)
2091.650066	**Equals L**	**in micrometers**

4C

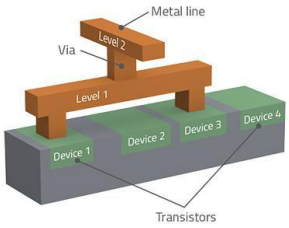
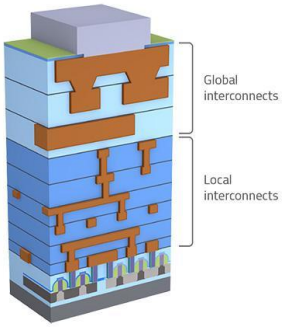
L	R/unit length	C/unit length
500	0.08	0.2
2000	0.08	0.2
5000	0.08	0.2
micrometer	ohms/micrometer	fF/micrometer

L	C/unit length	CL
500	0.2	100
2000	0.2	400
5000	0.2	1000
micrometer	fF/micrometer	fF

Pdyn = activity factor *C/L*L*Vdd^2*f

L	Cap. per unit length	Vdd	Activity factor	Freq	**Pdyn**
500	0.2	0.9	0.1	1	**8.1**
2000	0.2	0.9	0.1	1	**32.4**
5000	0.2	0.9	0.1	1	**81**
micrometer	fF/micrometer	V		GHz	microwatts

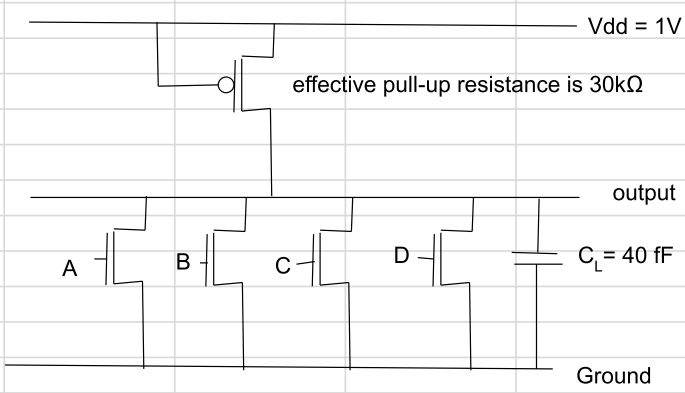
Image from: https://newsroom.lamresearch.com/Tech-Brief-An-Introduction-to-Interconnects

4D	 	
	Local Interconnect refers to thin, closely packed wire connections that connect close-proximity transistors. Global Interconnect refers to the thicker, longer wires that connect different areas of the chip and are further apart. As technology improves (ie as the transistors are able to be more closely packed together and reduce in size) the length and width of local interconnect wires also reduce. However, because global interconnect wires remain exceptionally long, spanning the chip even, their length does not decrease in the same way.	
	Information from:	https://newsroom.lamresearch.com/Tech-Brief-An-Introduction-to-Interconnects
4E	Repeater insertion would increase propagation delay, along with costs and area usage.	
	Using a wider wire would increase costs, area consumption, and the possibility of crosstalk between wires	

5A

Pseudo-NMOS NOR with one always-on PMOS pull-up load and four NMOS pull-down devices in parallel

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V ol = Voltage (output low)

Vol = Vdd(Rn/(Rp+Rn))

One NMOS pulldown on:

1*(10/(30+10))	equals Vol
0.25	**V**

4 NMOS pulldown on:

Resistance (total) = 10kOhms/4 because they are in parallel.

2.5	Ohms
1*(2.5/(30+2.5))	equals V ol
0.07692307692	**V**

Vol is input pattern dependent, because it depends on the resistance of the pull down network which is input-pattern dependent.

5B

NMOS ON	Voltage OL	R	
1	0.25	40	(30+10)
4	0.07692307692	32.5	(30+2.5)
	V	kOhms	

P=IV I=V/R ---> P=(V^2)/R

NMOS ON	V	R	P	**P**
1	1	40000	0.000025	**25**
4	1	32500	0.000030769230	**30.7**
		Ohms	Watts	**MicroWatts**

Falling output delay:

Pseudo NMOS does not output 0 ever. Shifts the falling output delay?

	Tau PHL =						
	300/40	7.5	Req in kOhms		ln ((vdd-vol)/(0.5-vol))		
	Vdd = 1	Vol = 0.25	---->	ln(0.75/0.25)			
	CMOS:						
	Tau PHL =	ln2* Rfalling *Cl					
	Req	10kOhms					
	Cl	40fF					
	Type	NMOS ON	Voltage OL	**Time**	**(for 1 NMOS on)**		
	Pseudo NMOS	1	0.25	**329.5836866**	**ps**		
5C	Static CMOS	1	0	**277.2588722**	**ps**		
	Rising output delay:		(no NMOS on)				
	Static cmos:	Worst case pull up resistance is 40kOhms					
	time =	Cl*resistance*ln2					
	time (ps) =	**1109.035489**	**for static cmos**				
	pseudo nmos:	Pull up resistance is 30kOhms					
	time =	Cl*resistance from 0.25V to 1V*(time factor)			ln ((vdd-vol)/(vdd-0.5))		
	time (ps) =	**486.5581297**	**for pseudo nmos**		equals Ln 1.5?		
	For a low power design, I would use the static CMOS design due to the pseudo-nmos' need for an always-on PMOS.						
5D	For a reduced transistor count, the pseudo NMOS design would be better, as there is only one PMOS transistor instead of 4 in series as required by the CMOS design.						
	For a reduced input capacitance, PMOS gates usually require more capacitance driving them. Reducing the number of PMOS gates is beneficial for this scenario, and so the pseudo nmos transistor would be preferred.						
5E							
	This would be bad due to the large string (6 of them!!) of PMOS transistors in series causing incredibly large resistance and delays. A better solution would be to separate the function using De Morgan, perhaps by splitting the inputs into two.						
	NOR ((NAND A,B,C),(NAND D,E,F)) maybe??						Ask yang wu